

Type 1 diabetes

What is Type 1 diabetes?

Type 1 diabetes is an autoimmune condition where the body's own immune system is activated to destroy the beta cells in the pancreas which produce insulin¹. We do not know what causes this autoimmune reaction however environmental factors are thought to set off the process. Type 1 diabetes is not linked to modifiable lifestyle factors. Currently there is no cure and it is lifelong.

Type 1 diabetes:

- Occurs when the pancreas does not produce insulin
- Represents around 10 per cent of all cases of diabetes and is one of the most common chronic childhood conditions
- In children, onset is usually abrupt and the symptoms obvious
- In adults, onset is slower
- Symptoms can include excessive thirst and urination, unexplained weight loss, weakness and fatigue and blurred vision
- Is managed with insulin injections several times a day or the use of an insulin pump

What happens to the pancreas?

In type 1 diabetes, the pancreas, a large gland behind the stomach, stops making insulin because the beta cells that make the insulin have been destroyed by the body's own immune system. Without insulin, glucose (a type of sugar) cannot enter the body's cells where it is usually turned into energy.

People with type 1 diabetes depend on [insulin](#) every day of their lives to replace insulin the body cannot produce. They must monitor their [glucose levels](#)

throughout the day to ensure they stay within their target glucose range. This can be done using a blood glucose monitor or continuous glucose monitor.

The onset of type 1 diabetes occurs most frequently in people under 30 years, however you can develop type 1 diabetes at any age. About 10 per cent of all cases of diabetes are type 1.

What happens if people with type 1 diabetes don't receive insulin?

Without insulin glucose builds up in the body. The kidneys attempt to wash the excess glucose out through the urine, resulting in dehydration. The body burns its own fat reserves to supply energy which releases chemical substances in the blood called ketone bodies. Without ongoing injections of insulin, these ketone bodies accumulate. The high glucose and ketone levels along with dehydration can be life threatening if it is not treated. This is a condition called diabetic ketoacidosis (DKA).

What causes type 1 diabetes?

The exact cause of type 1 diabetes is not known, but we do know it has a strong genetic link and cannot be prevented. People who develop diabetes may have one or a number of genes which make type 1 diabetes more likely, then some sort of environmental trigger occurs to start the autoimmune reaction. Examples of triggers include an infection or a high level of stress. Triggers are not always easy to identify.

Your doctor is likely to do a blood test to check you have auto-antibodies to confirm the diagnosis of type 1 diabetes.

We know that type 1 diabetes has nothing to do with lifestyle, although maintaining a healthy lifestyle is very important in helping to manage all types of diabetes, including type 1 diabetes.

At this stage nothing can be done to prevent or cure type 1 diabetes.

Symptoms of type 1 diabetes

- Being excessively thirsty
- Passing more urine

- Feeling tired and lethargic
- Always feeling hungry
- Having cuts that heal slowly
- Itching, skin infections
- Blurred vision
- Unexplained weight loss
- Mood swings
- Headaches
- Feeling dizzy
- Leg cramps

These symptoms may occur suddenly. If they occur, see a doctor. Through a simple test, a doctor can find out if they're the result of type 1 diabetes.

Untreated type 1 diabetes results in diabetic ketoacidosis (DKA), a medical emergency.

Management, care and treatment

Type 1 diabetes is managed with insulinⁱⁱⁱ injections several times a day or the use of an insulin pump. Insulin can't be taken as a tablet as it would be digested down to its building blocks, amino acids, by the body's normal digestive process.

Glucose levels can be measured using a blood glucose monitor or a continuous glucose monitor which measures glucose in the body's interstitial fluid^{iv}.

Following Australia's national guidelines on [healthy eating](#)^v and [exercise](#)^{vi} applies to all Australians, including people living with type 1 diabetes. Learning to recognise and [count carbohydrate](#) will assist with your accurate dosing of insulin.

If you are very newly diagnosed with type 1 diabetes there is often a 'honeymoon period' after you begin taking insulin. This is where your body's pancreas still produces some insulin, so your insulin needs may vary for a while. Eventually your pancreas will stop producing insulin completely. Your doctor may check how much insulin you are producing by ordering a laboratory test of C-peptide.

Having a [diabetes healthcare team](#) – a number of different health professionals – who you can turn to for advice, support and treatment is recommended.

These will include your general practitioner, endocrinologist (a specialist in hormones and diabetes), dietitian, diabetes educator, podiatrist, exercise physiologist, and psychologist, among others.

Having a regular healthcare routine for preventative checks with your healthcare team is recommended. Read about the diabetes annual cycle of care.

If you have recently been diagnosed with type 1 diabetes or have a family member with type 1 diabetes, view the information on [managing type 1 diabetes](#).

[i] Leszek Szablewski. (2011). *Glucose homeostasis and insulin resistance*. Bentham Science. <https://benthambooks.com/book/9781608051892/> p121-124.

[ii] Commonwealth of Australia. (2011). *National Evidence-Based Clinical Care Guidelines for Type 1 Diabetes in Children, Adolescents and Adults*. <https://diabetessociety.com.au/documents/Type1guidelines14Nov2011.pdf>

[iii] Overland, J. (2019). Type 1 diabetes: The overarching principles of insulin therapy [Review of *Type 1 diabetes: The overarching principles of insulin therapy*]. *Australian Diabetes Educator*, 22(1).

[iv] ADS- *Living Evidence Guidelines in Diabetes*. (n.d.). [Diabetessociety.com.au](https://diabetessociety.com.au/living-guidelines.asp). <https://diabetessociety.com.au/living-guidelines.asp>, accessed 14/9/2022.

[v] <https://www.eatforhealth.gov.au/guidelines>, accessed 14/9/2022.

[vi] Australian Government Department of Health. (2021, May 7). *Physical Activity and Exercise Guidelines for All Australians*. Australian Government Department of Health. <https://www.health.gov.au/health-topics/physical-activity-and-exercise/physical-activity-and-exercise-guidelines-for-all-australians>, accessed 14/9/2022.

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